

COM6012 Scalable Machine Learning 2026 Assignment

Deadline: 13:00 Wednesday 6th May 2026

Please carefully read the assignment brief before starting to complete the assignment.

Release Status: in progress

[Q1 - 10 marks](#)

[Q2 - 10 marks](#)

Q3 10 marks TBC

Q4 10 marks TBC

An [FAQ](#) will be updated when questions arise about important clarifications or tips.

Assignment Brief

How and what to submit

A. Create a folder **YOUR_USERNAME-COM6012-2026** containing the following:

- 1) **AS_report.pdf**: A report in **PDF format containing answers (including all figures and tables) to ALL questions** at the root of the zipped folder (like readme.txt in the lab solutions). If an answer to a question is not found in this PDF file, you will lose the respective mark. The report should be concise. You may include appendices/references for additional information, but marking will focus on the main body of the report.
- 2) **Code, batch scripts, and output files (see the sample solutions to the lab exercises as examples)**: All files used to generate the answers to the individual questions in the report above, except the data, should be included. These files should be named properly, starting with the question number (separate files for the two questions): **for example**, your Python code file should be named as **Q1_code.py** and **Q2_code.py**, your HPC scripts as **Q1_script.sh** and **Q2_script.sh**, and your output files from the HPC cluster as **Q1_output.txt** and **Q2_output.txt** (and **Q1_C1_fig.jpg**, etc.). The **results must be generated on the HPC cluster, not your local machine. Figures must be created by Python code.** We will apply a **25% penalty against the mark for a question for each missing file (code, batch script, output file, report).**

Example folder structure:

```
Shell
```

```
YOURUSERNAME-COM6012-2026/
```

```
|
|— AS_report.pdf
|
|— Q1/
|   |— Q1_code.py
|   |— Q1_script.sh
|   |— Q1_output.txt
|   └─ Q1_fig1.png
|
|— Q2/
|   |— Q2_code.py
|   |— Q2_script.sh
|   └─ Q2_output.txt
|
|— Q3/
|   |— Q3_code.py
|   └─ ...
|
|— Q4/
|   |— Q4_code.py
|   └─ ...
|
└─ ...
```

B. When you have finished ALL of the questions, zip your folder

YOUR_USERNAME-COM6012-2026 to include all of the files described above (one single report plus code, script, and output files for all questions, properly named) and upload this **YOUR_USERNAME-COM6012-2026.zip** file to Blackboard before the deadline. Double-check that these files are included by downloading the zipped folder from Blackboard to another machine and opening it.

C. **NO DATA UPLOAD:** Please do not upload the data files used. Instead, use the relative file path in your code, assuming data files are downloaded (and unzipped if needed) **under the folder 'Data'**, as in the lab.

D. **Code and output:** 1) Use **PySpark 4.1.0** and **Python 3.13** as covered in the lecture and lab sessions to complete the tasks; 2) **Submit your PySpark job to HPC** with **sbatch** to obtain the output. Output files must be produced by the HPC batch jobs submitted via sbatch.

Assessment Criteria (Scope: Sessions 1 to 8; Total: 40 marks - 10 marks/question)

1. Being able to use PySpark to analyse big data to answer data analytics questions.

2. Being able to perform tasks covered in Sessions 1 to 8 on large-scale data.
3. Being able to make useful observations and clearly explain the obtained results.

Late submissions: We follow the university's guidelines about late submissions, i.e., "If you submit a piece of coursework late, your mark will be deducted by **5% for each working day after the deadline**.

If you submit more than five working days after the deadline, **you will receive a mark of zero.**" Please see [this link](#).

Extenuating circumstances: If you are experiencing difficulties that are affecting your assignment submission, you may be able to apply for extenuating circumstances so these difficulties can be taken into account. Please see [this link](#).

Academic misconduct: "The consequences of academic misconduct can be serious. You may face action within your school or at a University discipline hearing." Please carefully read [this link](#) on different types of Academic Misconduct and how to avoid them if you are not sure.